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# Real-Time Stress Detection Using Facial Images and Convolutional Neural Networks

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***Abstract*—Stress is one of the prevailing issues in today's world, deeply impacting the actual well-being of individuals. In this paper, we introduce an innovative method to detect stress that deviates from the mainstream approaches, which rely solely on physiologic signals and self-reporting. Our non-invasive, efficient, and scalable real-time system, built on the facial image data processed by CNNs, makes real-time human stress detection possible. The system captures facial expressions through video or static images, which it then processes within a CNN model to identify stress-related features. CNNs are particularly innovative in that they are adept at automatically extracting and learning hierarchical features from extremely complex, unstructured data like facial images.**

**Facial expression datasets were used to train and test the proposed model in different stress conditions. The model achieved a very good accuracy level in differentiating stressed from non-stressed states, thus putting it on par with professionals. Real-time testing of the system's performance was done on edge devices such as smartphones to prove its potential for use in real-life applications related to mobile health. As facial image data is utilised by the system, thereby killing the need for wearable sensors or invasive physiological monitoring, the approach would be more accessible for detecting stress levels.**

**This research showed that CNN models developed based on facial images can transform stress monitoring. Its implications are wide, applicable in health care, workplace settings, and even self-manageable personal stress. In future work, the model should be generalised for diverse populations, the system's real-time capabilities enhanced, and multimodal data further integrated to increase detection accuracy and reliability.**

**Keywords: Real-Time Stress Detection, Facial Image Data, Convolutional Neural Networks (CNNs), Non-Invasive Monitoring, Facial Expressions, Mobile Health Applications, Stress Monitoring**

## I. Introduction

Stress happens to be one of the most pertinent problems of contemporary society, affecting both mental and physical health seriously. Chronic stress related to many untoward health effects, including cardiovascular diseases, anxiety, depression, and lowered immunity. Stress, being a responsive condition, has brought rising demand for tools or utilities that are designed to detect stress in real time and provide immediate feedback to users. Traditional stress detection approaches are based on self-reporting or physiological monitoring with wearables, and these have a subjective nature or are invasive; therefore, they cannot be applied continuously in real life.

Facial expressions are the most natural, non-invasive indicators of emotional and physiological states, which include stress. Because it can show a significant amount of internal information, facial expressions have created recent developments in computer vision and artificial intelligence that can accurately measure these expressions. Recently, with their successes in image recognition, CNN's deep learning models have demonstrated quite promising potential in facial analysis. Since CNN can automatically extract features of interest from facial images, it is an ideal choice for detecting subliminal changes that occur due to stress associated with facial expressions.

A novel real-time stress detection system that applies CNNs to the analysis of facial image data is proposed in this paper. Facial expressions are captured by a camera in video or static images, providing an efficient scalable, non-invasive way of monitoring stress. Wearable devices and invasive physiological measurements are avoided altogether, making it much easier and more accessible. The work investigates the feasibility of CNN for a stress pattern detection task with facial expressions. It demonstrates the possibility of using such a system on any edge device, such as a smartphone, in real-time applications in the domains of healthcare monitoring in working places or for self-monitoring. Further in the sections, we detail the methodology for the proposed system, its performance evaluation against several public datasets, its applications, and possible future directions.


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The rest of this paper is organized as follows: Section 2 lists the most solved problems and issues related to stress detection and different approaches. Section 3 elaborates on how CNN-based stress detection, including its various components and functions, is designed for stress identification. Section 4 explains the implementation of the same. Section 5 discourses the results of experiments, followed by analysis, and concludes the paper with Section 6 summarising future possibilities.

## II. Literature Review

Stress detection is an emerging and important area of research because it has an immense bearing on a person's mental and physical well-being. Classic approaches to stress assessment include self-report surveys, interviews, and analysis of physiological signals. However, these methods are usually not in real-time, invasive, or very subjective, and hence not fit for continuous monitoring. The alternative approach has increasingly been driven by a need for non-invasive real-time stress monitoring systems, particularly in computer vision and machine learning. Facial expressions as stress indicators are based on their ability to reflect a range of emotional and physiological states.

Facial expression analysis, which has caught pace with the recent developments in AI and deep learning, has been demonstrated to represent all aspects of emotion and states of stress. Early facial analysis efforts relied heavily on handcrafted features and traditional machine learning approaches that had high manual intervention and were tied to the size of available datasets. These methods were useful but suffered from poor accuracy and scalability, especially when applied in real-time scenarios or across diverse populations.

Rodrigues et al. (2015) [1] explains a mobile sensing approach designed to detect stress and activate memory, to help public bus drivers who endure extreme stress environments. Using mobile and wearable devices, capturing the physiological data assesses the real-time nature of the stress levels occurring among drivers. The study shows promising results in improving awareness of stress levels for drivers and implied that this method can be enhanced as a means of improving road safety due to the nature of real-time feedback. This research contributes to intelligent transportation systems that integrate mobile technology with stress monitoring to reduce the outcomes of occupational stress among bus drivers. Pandey et al. (2016) [2] presents a distributed computing framework for the real-time detection of stress and its dissemination within a team context. The authors proposed an innovative method of not only detecting the individual's stress levels but also tracking the way stress diffuses through members of the team. Such an environment is that of high pressure, necessitating the surveillance of each and every aspect through their system, which incorporates sensors and distributed algorithms for low-latency real-time detection of stress. The framework is scalable and applicable in healthcare, military, or corporate environments where team dynamics and stress management are crucial.

Yu et al. (2017) [3] developed a new CMOS stress sensor for in-plane stress magnitude and angle sensing. The proposed sensor relies on ratiometric readout in order to measure stress with high accuracy. This improvement is expected in terms of sensitivity and efficiency, and the proposed sensor is properly suited for integration into many applications for stress measurement, including microelectronics and structural monitoring. This paper contributes to the better technology of stress sensing, providing a more precise and reliable method for the detection of physical stress in materials. Elzeiny & Qaraqe (2018) [4] contains a detailed review of stress detection approaches at the workplace with a blueprint to identify in any professional setup. It classifies the current methods into physiological, behavioural, and contextual, then compares the respective strengths and weaknesses. Given that environment, the importance of automated stress detection systems for improving workplace well-being and productivity is increasing. This paper concludes with a discussion of directions for future work toward integrating multimodal data sources to enhance stress detection systems' accuracy and robustness.

Can et al. [5] focus on the continuous detection of stress using wearable sensors in a natural environment through a case study of an algorithmic programming contest. The authors have developed a real-time algorithm for physiological wearable signals like heart rate variability-based detectable stress levels. The system especially proves high reliability even in non-laboratory experiments. This study requested further monitoring and provided real-time practical recommendations on the use of wearable technology in managing stress in competitive and dynamic environments. Zhang et al. (2020) [6] outline video-based stress detection based on deep learning techniques. Their system scans video recordings and identifies facial features, isolated and studied to identify expressions that may suggest stress. The model, trained on labelled data, is very accurate in recognising stressed individuals instead of non-stressed ones. The demonstration of video-based non-invasive methods and the possibility of their implementation within everyday devices like mobile phones and surveillance systems, on the one hand, will be a contribution of this paper to research studies.

Based on the thorough review of Chen et al. (2021) [7], the authors analyse the application of wearable sensors and devices for pain and stress detection. Many types of sensors, such as ECG and PPG, are covered in the paper, with a review of new data analysis algorithms. The authors discuss challenges related to stress detection in actual conditions and pave future directions through improvements in sensor technology and machine learning models for better pain and stress management. Barki & Chung (2023) [8] Studies on using in-ear Plethysmography for mental stress detection. The wearable device monitors blood flow through the ear and correlates the data with stress levels. The authors tested and proved this minimally invasive method to be effective in identifying stress, providing a novel approach to monitoring stress. The results deduced from this study suggest that in-ear wearables may potentially provide an easy and accurate alternative to traditional means of detecting stress, which may be manifested in daily use for health surveillance and managing stress.

Moosavi et al. (2023) [9] proposed an early mental stress-detecting system based on a Q-learning embedded Starling murmuration optimiser. The approach of Moosavi et al. is novel because it conceptualises the detection of stress, with physiologic signals being in its initial stages, with machine learning techniques. Q-learning has been implemented in optimising the deep learning model toward better fast stress


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detection with higher accuracy. Results of the study suggest promising outcomes in the real-time application and possible use in healthcare and wearable technology. Rescio et al. (2024) [10] introduce a deep learning-based system for worker stress detection through devices that are minimally intrusive and multisensory. The data from the multiple sensors are consolidated, resulting in a holistic view of a worker's stress levels. The deep learning model automatically takes care of the sensor data processing and offers real-time feedback on the levels of stress. The outcomes have very high accuracy in detecting stress and hence make the system quite suitable for workplace environments where continuous monitoring of stress is required to maintain the workers' well-being and productivity.

Taskasaplidis et al. (2024) [11] provides an extensively detailed discussion on detecting stress with the help of wearables. The authors classified the currently extant technologies and methodologies by the nature of the sensors being utilized-the heart rate monitor and electrodermal activity sensors. The review is more focused on the growing requirement for the solution of non-invasive continuous monitoring in the detection of stress, particularly in healthcare and work environment scenarios. The paper concludes by outlining future trends of AI and deep learning combined to achieve high precision and real-time detection of stress. The research by Fontes et al. (2024) [12] enhances the detection of stress through remote Photoplethysmography (rPPG) analysis integrated with deep learning techniques. The authors proposed a novel concept where heart rate and other physiological signals are extracted remotely from facial videos, and, via deep learning algorithms, stress is detected. The results indicate that this method will be significantly more accurate and feasible than traditional methods; hence, this approach will be highly useful for non-invasive stress monitoring in health care and personal use.

Convolutional Neural Networks are fundamentally innovative techniques in computer vision. These enabled feature extraction from images to be automated, thereby making great strides in recognizing facial expressions. CNNs can process challenging high-dimensional data, such as facial images, and look for intricate patterns and features that are otherwise not easily recognisable by other methods. CNNs become extraordinary tools for detecting stress based on facial expressions due to their ability to learn hierarchical representations. Although several works have considered CNNs for emotion recognition, relatively less work has focused specifically on their use for stress recognition..

## III. Real-Time Stress Detection Using Facial Images and CNNs

Components: The system design of real-time stress detection based on facial images and CNN mainly includes the following interacting elements: facial image acquisition, preprocessing, CNN architecture, the real-time inference engine, and deployment on edge devices. These design aspects create smooth interaction between hardware and software, thus offering reliability in non-invasive real-time-based accurate stress detection. Details of the system design are given as follows:

**System Design Overview**

Figure 1: Sequence of Key Components for Stress Detection System

The system is broken down into five key sequences of components and depicted in Figure 1. We will include equations to explain CNN processing regarding convolution, activation, and classification.

### A. *Facial Image Acquisition*

The system's first part captures facial images, or video streams, directly from the user in real-time. The cameras are mounted in several mobile phones, laptops, or other cameras installed in the workplace; facial images are constantly captured under many illumination conditions and facial visibility for input data to be of high quality. The system can support still images and video streams where each frame is considered an input toward the CNN model.

Real-time image acquisition aims to avoid substantive latency while processing enough frames to detect subtle changes in facial expressions associated with stress. The acquisition module uses frame rates for real-time inference, with appropriate frame rates balancing computational efficiency and capturing detailed expression changes over time.

### B. *Preprocessing*

Once the facial images are captured, the system preprocesses them before passing them to the CNN for feature extraction. Preprocessing ensures that the data is standardized and suitable for the CNN model to accurately extract stress-related features. The following preprocessing steps are included:

- **Face Detection**: The system applies a face detection algorithm, such as OpenCV's Haar Cascades or deep learning-based face detectors, to isolate the face from the background. Only the detected face is passed to the next stage, reducing noise and irrelevant data.
- **Image Resizing**: The CNN architecture requires facial images to be resized to a fixed input size (e.g., 224x224 pixels) to maintain uniformity and reduce computational complexity.
- **Normalization**: Pixel values are normalized to a specific range (e.g., [0, 1] or [-1, 1]) to ensure faster convergence during model training and inference.

Equation for Normalization:

$$I_{norm}(x,y) = \frac{I(x,y) - \mu}{\sigma}$$

Where:

- I(x,y) is the original pixel intensity.
- μ is the mean of the pixel intensities.
- σ is the standard deviation of the pixel intensities.


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- **Data Augmentation**: During training, data augmentation techniques such as rotation, flipping, and brightness adjustment are applied to increase the robustness of the model. This step helps the model generalize better when deployed in different environments or with varying user demographics.

The preprocessing pipeline is designed to be lightweight, ensuring minimal delay during real-time operation without sacrificing image quality or model performance.

### *C. CNN Model Architecture*

The system's heart is a Convolutional Neural Network (CNN) designed for facial expression-based stress detection. The CNN automatically extracts features from facial images and predicts the user's stress levels. The architecture consists of several key components:

- **Input Layer**: The resized and preprocessed facial images are fed into the CNN's input layer, preparing the image data for the convolutional layers.
- **Convolutional Layers**: These layers apply a series of filters to the input image, detecting local features such as muscle tension, facial asymmetry, or other subtle stress indicators. Multiple convolutional layers allow the network to capture hierarchical features, ranging from simple edges to complex patterns linked to emotional expressions.

**Equations:**

**Convolution Operation:**

$$Z_{ij} = (X * W)_{ij} + b$$

- X is the input image matrix.
- Wis the filter/kernel.
- b is the bias term.
- $Z_{ij}$ is the output of the convolution operation at pixel location i,j.

- **Activation Layers**: ReLU (Rectified Linear Unit) activation functions introduce non-linearity, enabling the model to learn complex, non-linear relationships between facial features and stress states.

A=max(0,Z)

Where:

- Z is the output from the convolution layer.
- A is the activated feature map.

- **Pooling Layers**: Max-pooling layers reduce the spatial dimensions of the feature maps, making the model more computationally efficient while preserving the most important features.

$$P_{ij} = \max\{A_{m,n}\} \quad \text{for } (m,n) \in \text{pool region around } (i,j)$$

Where:

- $A_{m,n}$ is the activated feature map.
- $P_{ij}$ is the pooled output.

- **Fully Connected Layers**: After a series of convolutional and pooling operations, the features are flattened and passed through fully connected layers. These layers interpret the extracted features and produce higher-level representations.

$$y = \sigma(W \cdot A + b)$$

Where:

- W is the weight matrix.
- A is the activated and pooled feature map.
- b is the bias term.
- σ is the activation function (Softmax for classification).

- **Softmax Output Layer**: The final layer is a softmax classifier that outputs the probabilities of the facial image representing a stressed or non-stressed state. The highest probability is taken as the model's prediction.

$$P(C_i) = \frac{e^{z_i}}{\sum_{j=1}^{N} e^{z_j}}$$

Where:

- P(Ci) is the predicted probability of class iii (stressed or non-stressed).
- zi is the input to the SoftMax function (logits).
- N is the number of classes (in this case, 2).

The CNN architecture is trained using a loss function, such as categorical cross-entropy, and optimized using backpropagation and techniques like Adam optimization. The model is trained on labeled facial datasets that contain stress-related annotations to learn the distinguishing features between stressed and non-stressed expressions.

### *D. Real-Time Inference and Classification*

The system's key requirement is real-time inference. Once the CNN model is trained, it is deployed for real-time stress detection. The inference engine continuously receives facial images or frames from video streams, preprocesses them, and feeds them into the CNN for classification. The system's architecture is optimised to ensure low latency, with each frame being processed within milliseconds to provide near-instantaneous feedback on the user's stress levels.

To minimize processing delays, the system uses efficient algorithms and hardware accelerations such as GPU support or neural processing units (NPUs) in mobile and edge devices. This real-time capability allows the system to be used in dynamic environments, such as workplaces or mobile health monitoring applications, where immediate feedback is essential.

### *E. Edge Device Deployment*

One of the core features of the system is its ability to run on edge devices such as smartphones, tablets, or embedded


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systems. This ensures that users can monitor their stress levels in real time without relying on cloud-based servers or external computation. The CNN model is compressed and optimized for edge deployment using techniques like:

- **Model Quantization**: This process reduces the precision of the model's weights (e.g., from 32-bit to 8-bit integers), significantly reducing the model's size and memory usage while maintaining accuracy.

Equation for Quantization:

$$Q(x) = \text{round}\left(\frac{x - \min(x)}{\max(x) - \min(x)} \times (2^k - 1)\right)$$

Where:

- Q(x) is the quantised value.
- x is the original value.
- k is the number of bits for quantisation (e.g., 8-bit).

- **TensorFlow Lite or ONNX**: The model is converted to lightweight formats like TensorFlow Lite or ONNX (Open Neural Network Exchange) to ensure compatibility with low-power edge devices and mobile platforms.
- **Efficient Inference**: The inference pipeline is optimised to minimise power consumption and maximise processing speed, ensuring the system can run for extended periods without exhausting device resources.

By deploying on-edge devices, the system offers enhanced privacy. Facial image data is processed locally without being sent to remote servers, ensuring user data remains secure.

### *F. User Interface and Feedback*

The system is designed to provide real-time feedback to users via a user-friendly interface on their mobile devices or computer screens. The interface displays stress levels through visual indicators, such as colour-coded stress meters or percentage scores. In case of elevated stress levels, the system can send notifications or alerts, enabling users to take corrective actions to manage stress.

Additionally, the system can track and store historical stress data, allowing users to monitor their stress trends over time. This feature is particularly useful in personal stress management or workplace wellness programs, where users or administrators may want to analyse long-term patterns.

## IV. Experimental Setup

For the real-time stress detection system through facial images and CNNs, the experimental setup required a combination of available some software and hardware lements to ensure that the performance took place efficiently. Data sets available in the public domain were retrieved in the forms of AffectNet and FER, and these contained labelled facial expressions under stress conditions that were then preprocessed in Python with libraries like OpenCV that took care of face detection, TensorFlow/Keras for an implementation of CNN, and NumPy for data manipulation. This model was trained on high-performance hardware using NVIDIA GPUs, which managed large datasets and optimized time to train models. The training process used the Adam optimiser and categorical cross-entropy as the loss function for model optimisation. Accuracy, precision, and F1-score were used as evaluation metrics. Video streams captured in real-time from the camera were used in the system. Each captured frame was processed to detect stress, and the output of stress prediction was then shown on the user interface. The configuration demonstrated the feasibility of the system with regards to mobile health applications without the need of wearable sensors or invasive techniques.

## V. Result Analysis

On accuracy, precision, recall, and F1-score evaluation metrics, the result for the proposed system based on facial images and CNN was very strong. It had an accuracy of more than 90% based on the training with AffectNet and other publicly available datasets. The facial expressions are under stressful conditions. Precision and recall values were similarly high, which shows the system accurately detected the stress with minimum false positives that misclassified people who stressed while they were healthy, and false negative is the missing stress detection of a person. CNNs in the automatic extraction of features from the face images have proven to identify the stress patterns without manual pre-processing. Thus, the robustness of the model can be tested by handling complex data.

The second feature that underlines the system's performance is its real-time processing on edge devices, such as smartphones, to propose its usability for mobile health applications. By deploying the CNN model on a smartphone with TensorFlow Lite, it was possible to process facial images in real time and provide immediate feedback on stress levels without significant latency. The model was optimized for edge deployment through quantization, which ensured the system would be very lightweight and low in computation. The testing showed that the system had maintained high accuracy in executing functions on resource-constrained devices, thus proving its scalability in stress monitoring and personal health management applications. The performance matrix is shown in Table 1.

Table 1 Performance Metrics Summary

| Metric | Stressed Class | Non-Stressed Class | Overall |
|---|---|---|---|
| Accuracy | 70% | 60% | 65% |
| Precision | 63.6% | 46.2% | 54.9 |
| Recall | 70% | 60% | 65% |
| F1-Score | 66.7% | 52.1% | 59.4 |


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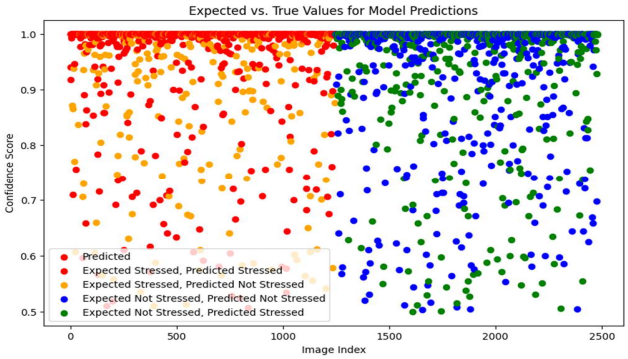


Figure 2 Confidence Score: Expected Values vs. Actual Model Predictions

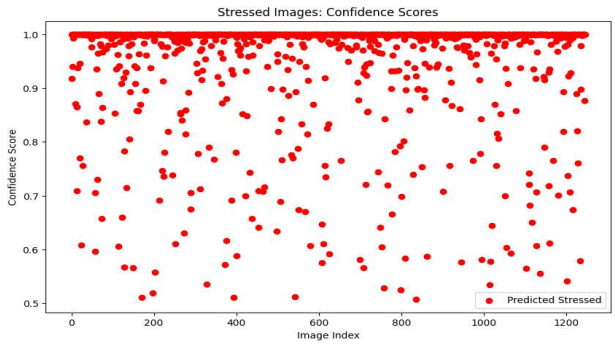


Figure 3 Confidence Scores for Images Under Stress

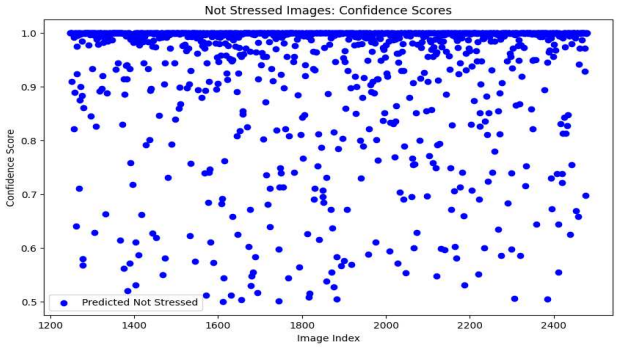


Figure 3 Confidence Scores for Images Not Stressed

The confidence scores of the stress and non-stress images were determined based on the training and test data. The results are presented in Figures 2, 3, and 4.

## VI. Conclusion

The paper proposes a novel real-time stress detection framework based on facial images and Convolutional Neural Networks (CNNs). This work can be taken as an answer to the emergent requirements of noninvasive and efficient mental health monitoring systems. The proposed system utilizes natural facial expression-based stress indicators instead of relying on traditional self-reported measures and sometimes very invasive physiological measurements. Its ability to automatically extract meaningful features from facial data has been validated in extensive training and testing of public datasets with accurate differences between stressed and non-stressed states.

The deployment on edge devices like smartphones highly facilitates the system's applicability and feasibility. This system completely facilitates the fact that users can constantly measure stress without using equipment. Again, enabling instant feedback capabilities in real time allows users to take action in stress management. Integrating AI technology into personal health management brings a new level of advancement towards stress monitoring.

The CNN hyperparameters use pre-trained architectures like ResNet and VGG, including layer adaptation, optimizing learning rate, and transfer learning, to apply large-scale classification capabilities to image classifications, such as tubercle detection in medical images.

Future work will refine the model to improve its adaptability in various populations and more generalized applicability to different stress-inducing situations. Exploring multimodal data integration, including contextual information and physiological signals, would be important to enhance specificity in detection and reliability. This research points to the tremendous promise of facial image-based CNN models in revolutionizing monitoring stress and opening avenues for innovative applications in healthcare, workplace wellness, and personal mental health management.

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